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Geometry & Mesh in Marc Mentat

Materiały i Struktury Inteligentne (2018) – przygotował: Jakub Bryła
opieka merytoryczna: dr inż. Adam Martowicz

Agenda:

-> Marc Mentat interface

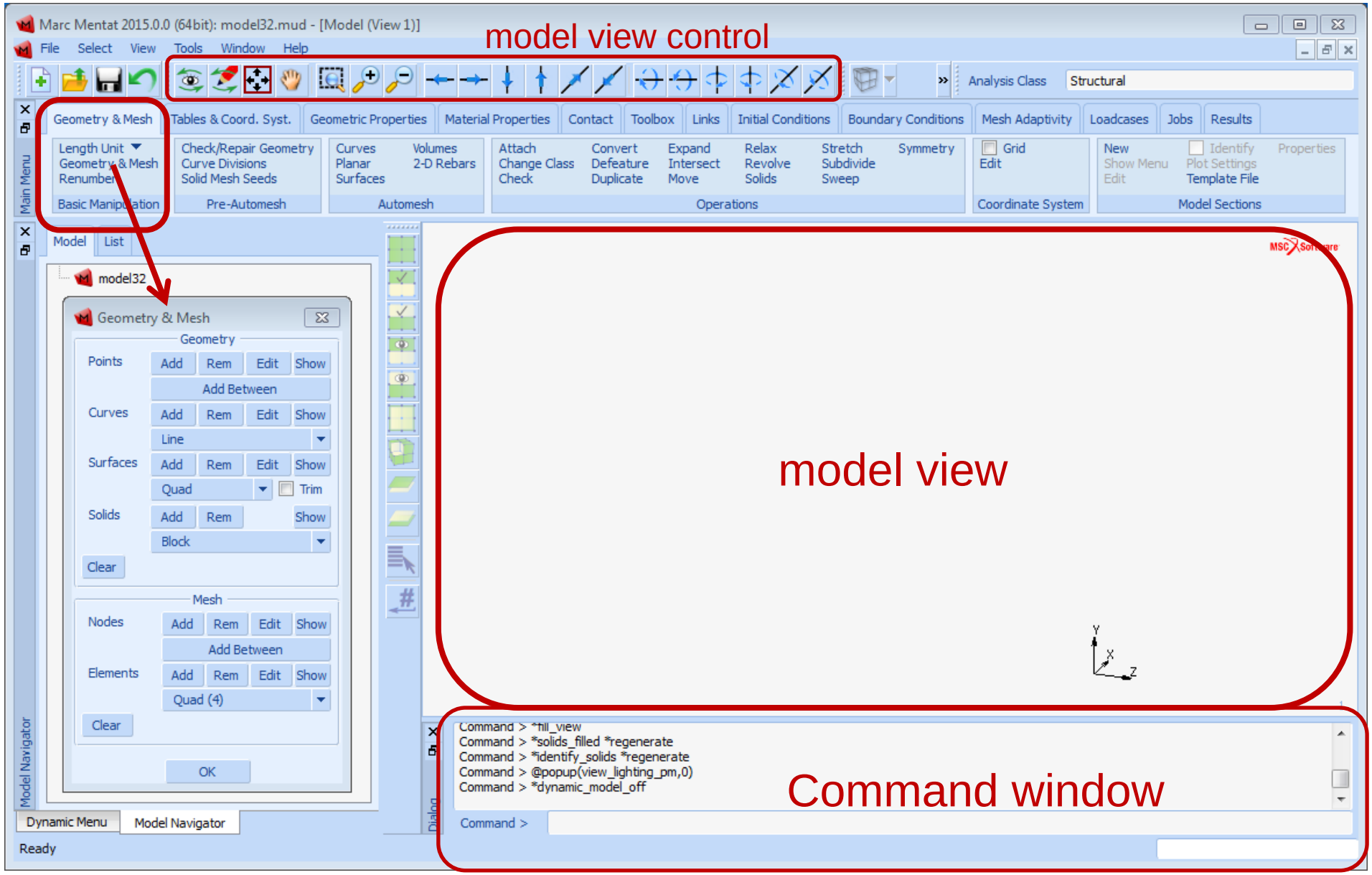
-> 1D modeling:
=> from geometry to mesh
=> direct mesh generation

-> 2D modeling:
=> from geometry to mesh
=> direct mesh generation

-> 3D modeling:
=> direct mesh generation
=> import geometry and mesh

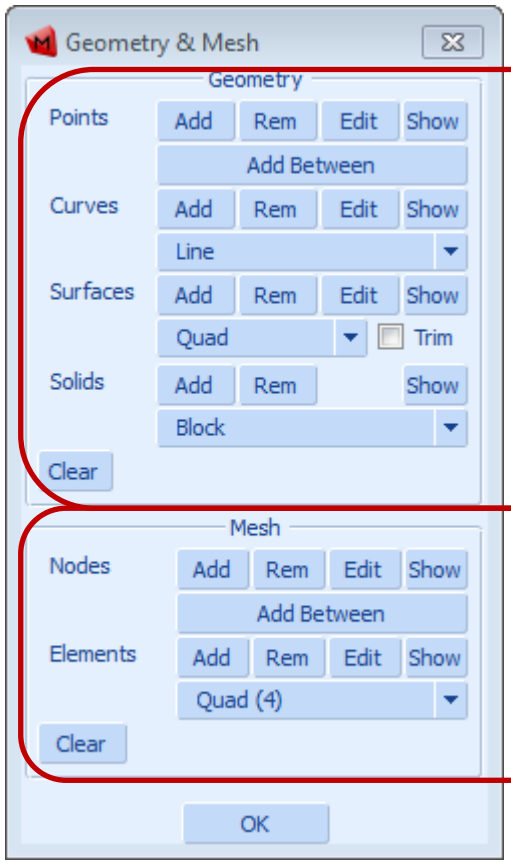
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Marc Mentat interface



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Marc Mentat interface



Menu supports creating the geometry.

The created geometry must be meshed in next step.

Menu supports creating the mesh item directly.

Such model is ready to analysis.

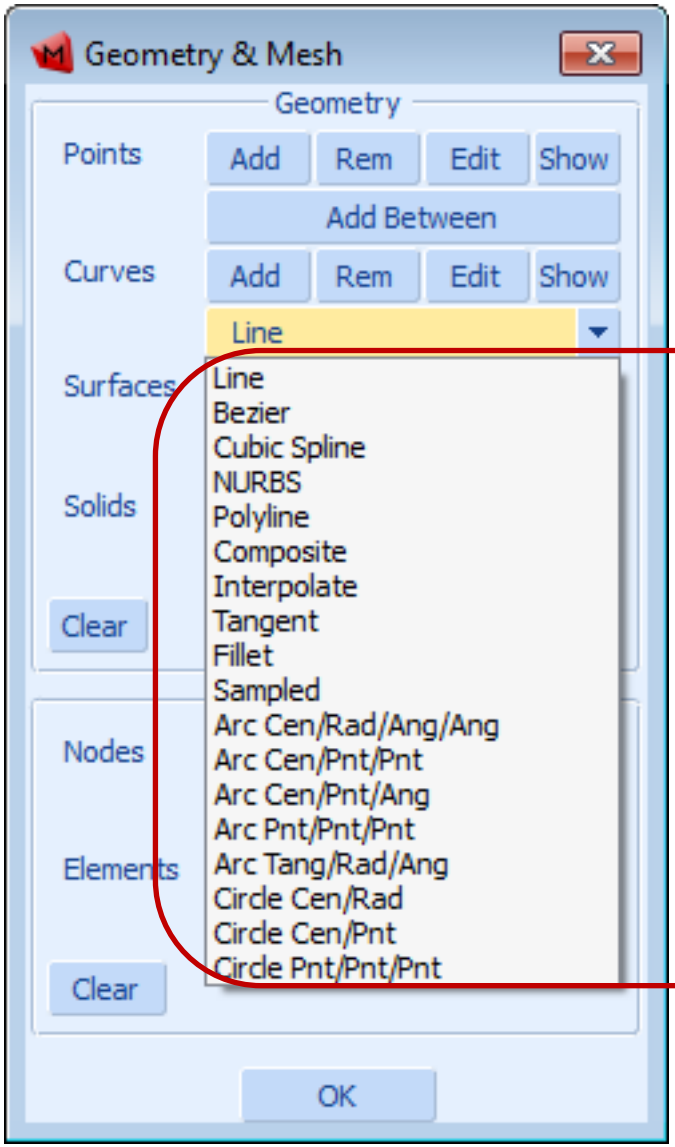
1D modeling:

=> **from geometry to mesh**

=> direct mesh generation

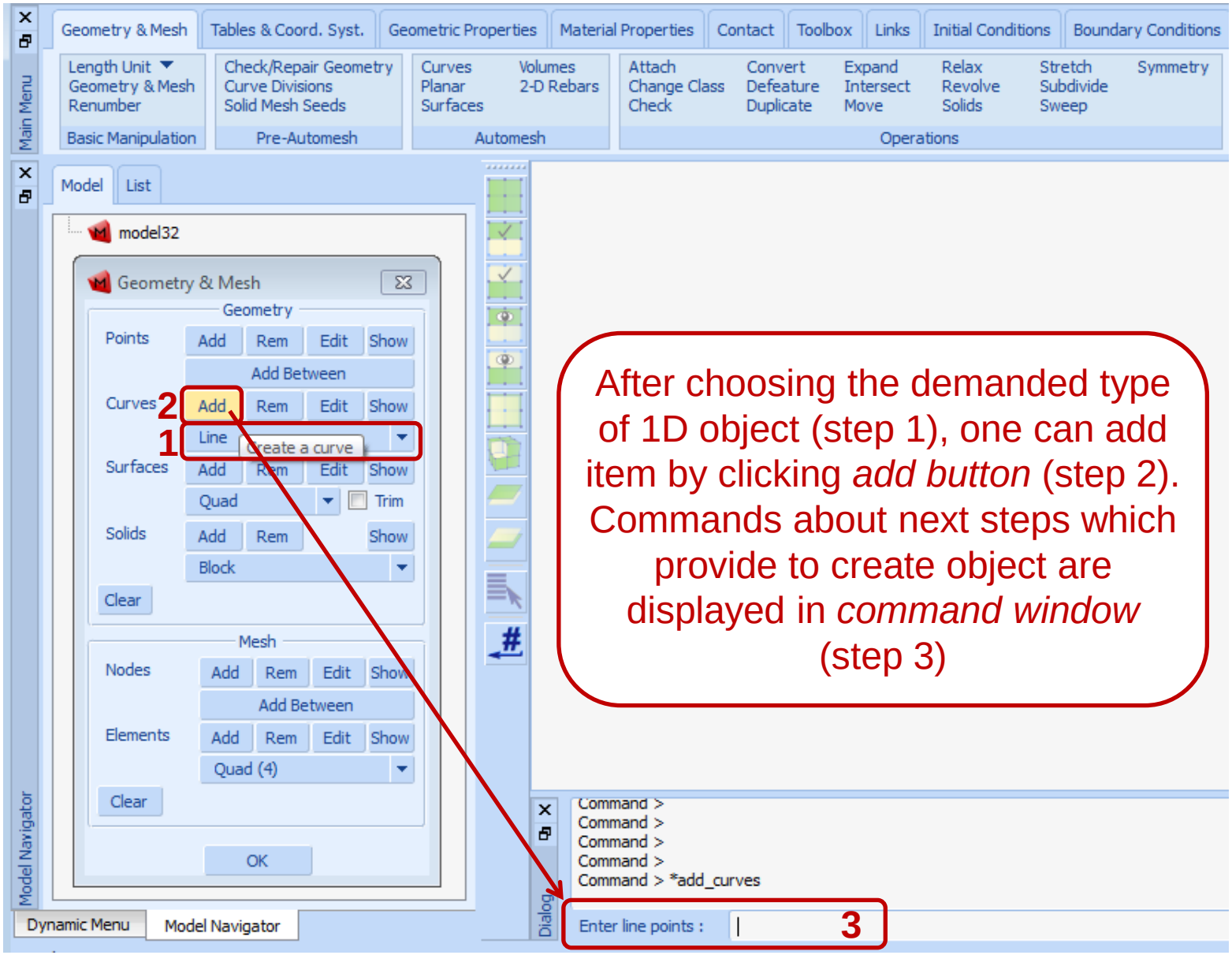
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Type of 1D objects

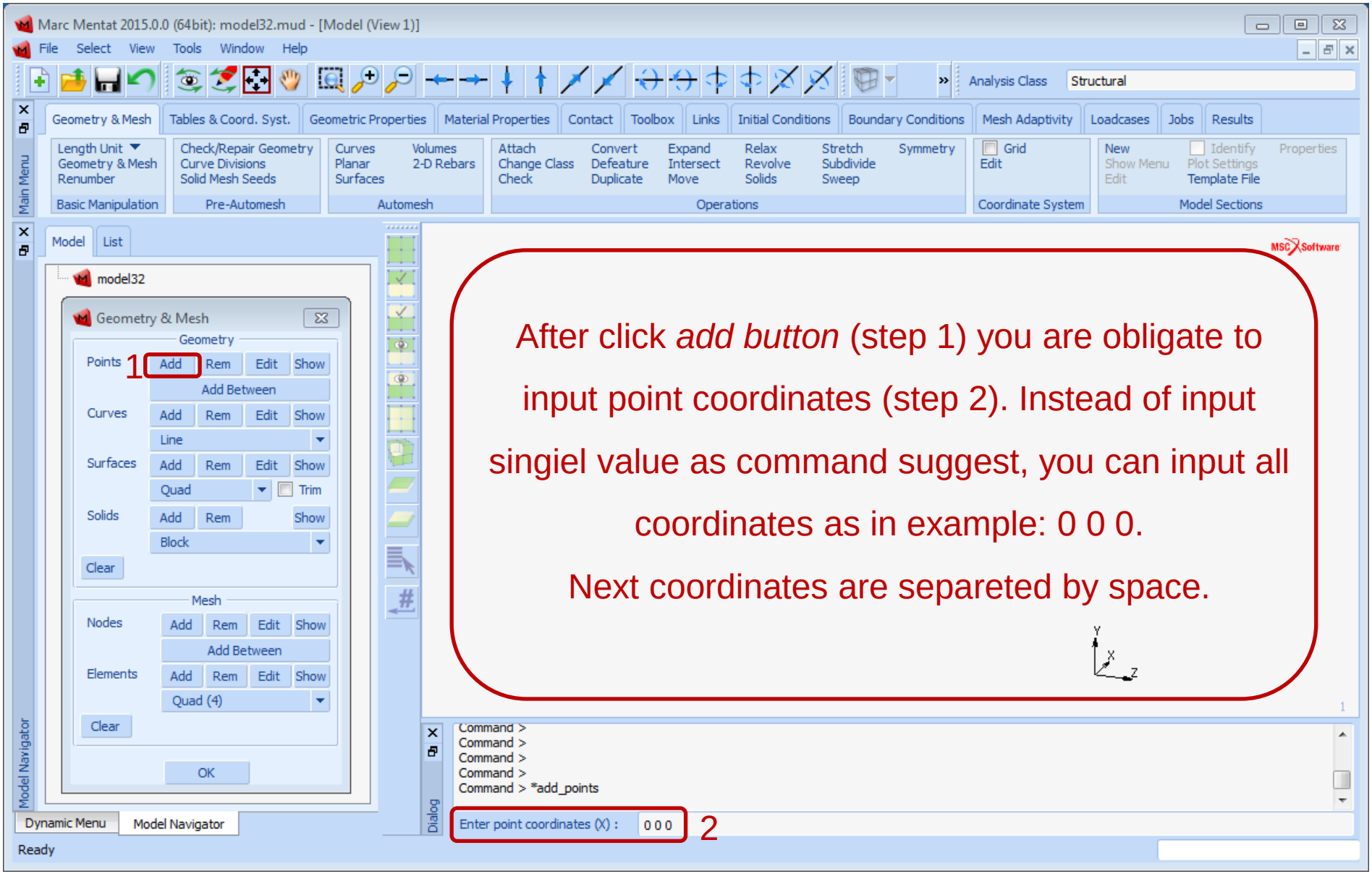


In more cases (or even each) the points must be defined for possibility to creating the curves objects.

Creating 1D objects

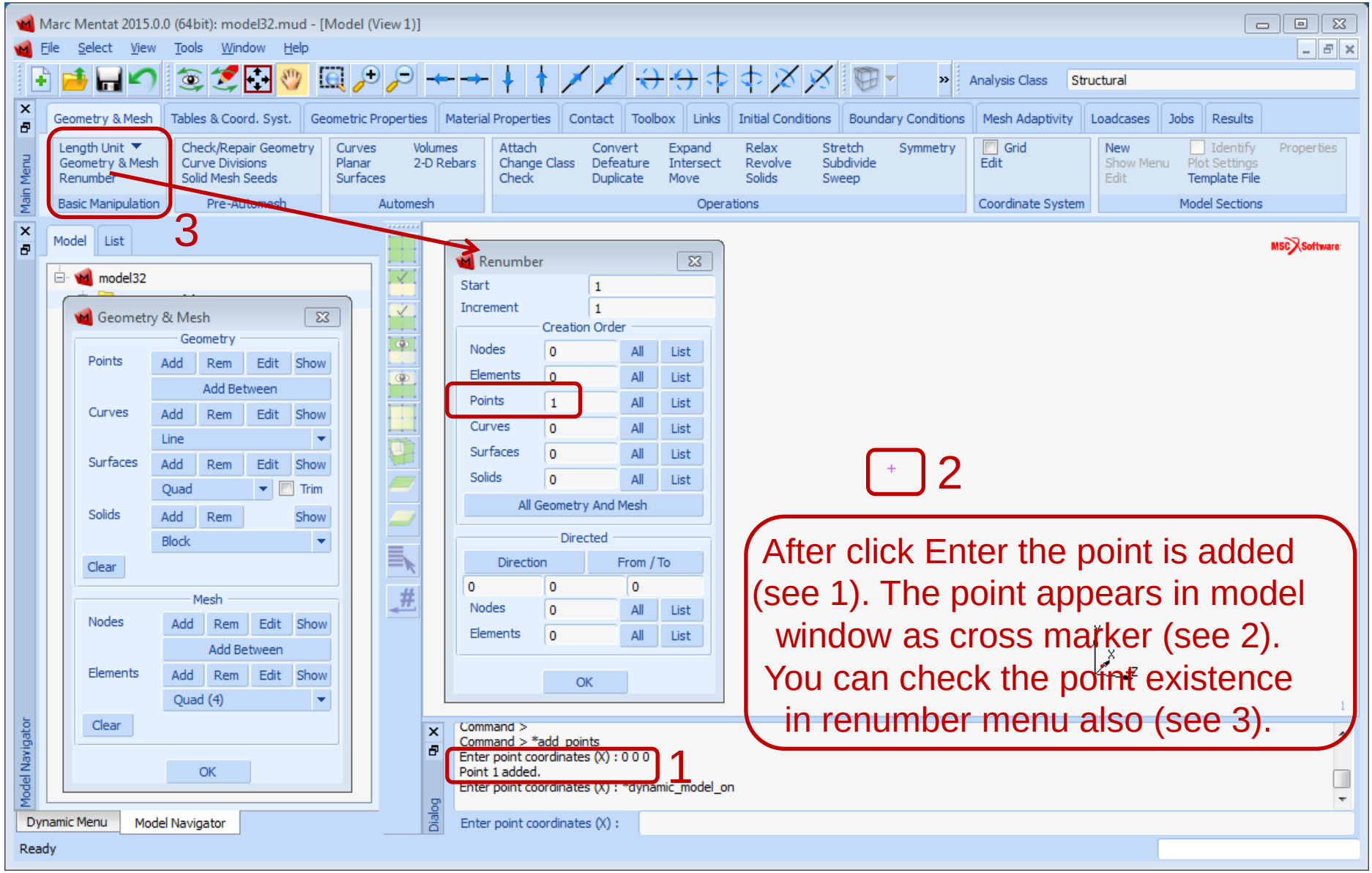


Creating 1D objects - example



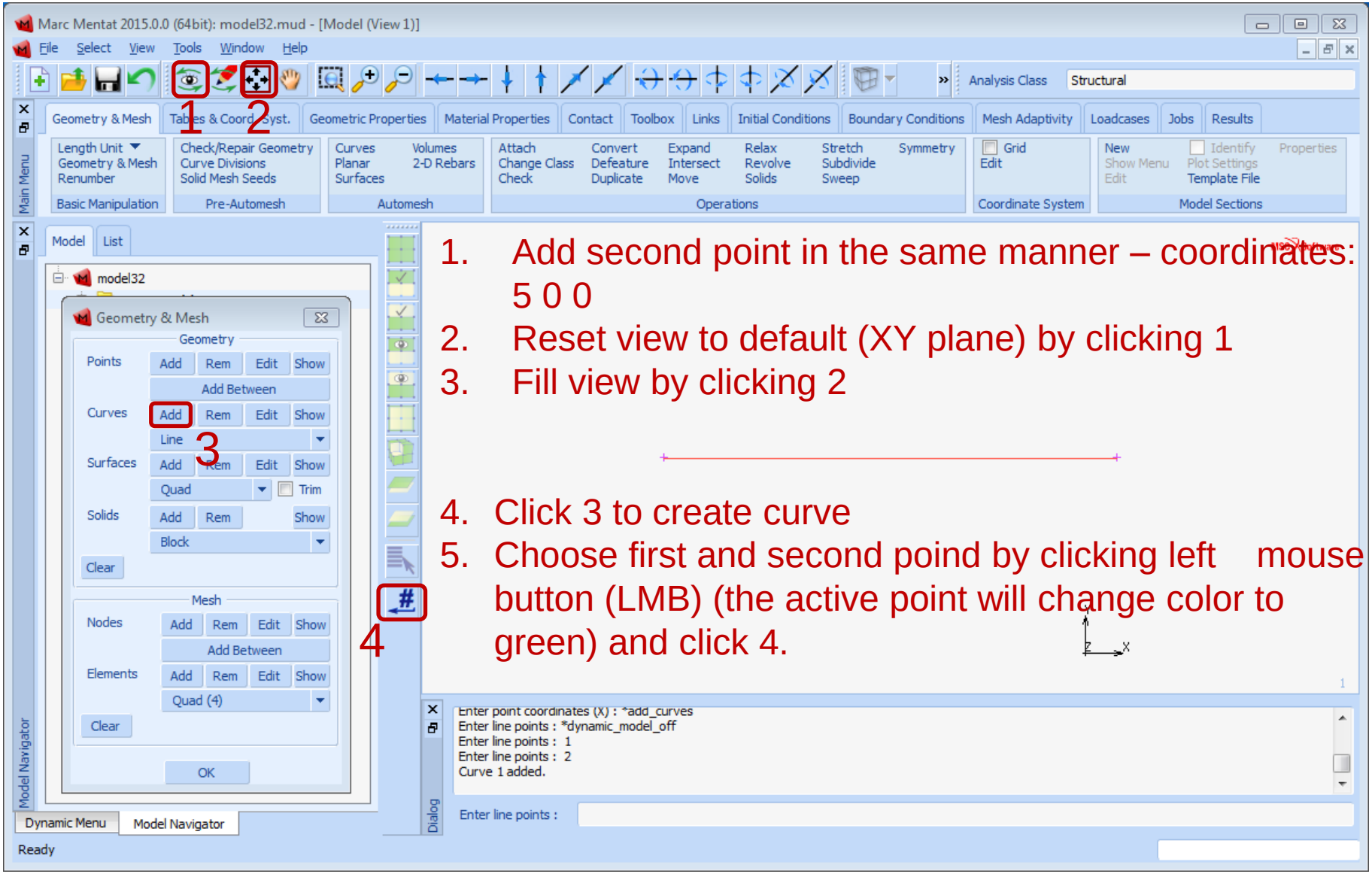
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Creating 1D objects - example



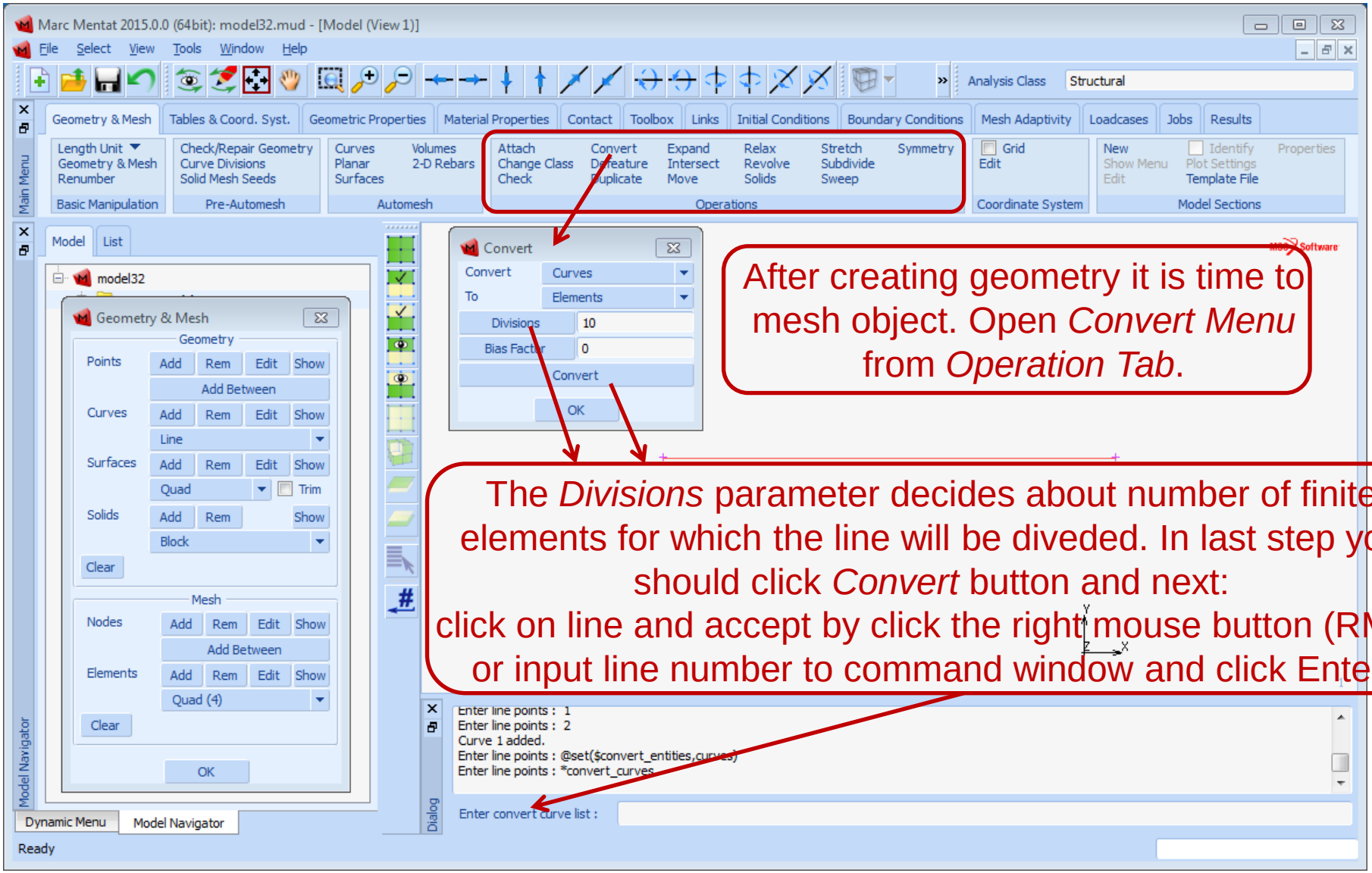
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Creating 1D objects - example



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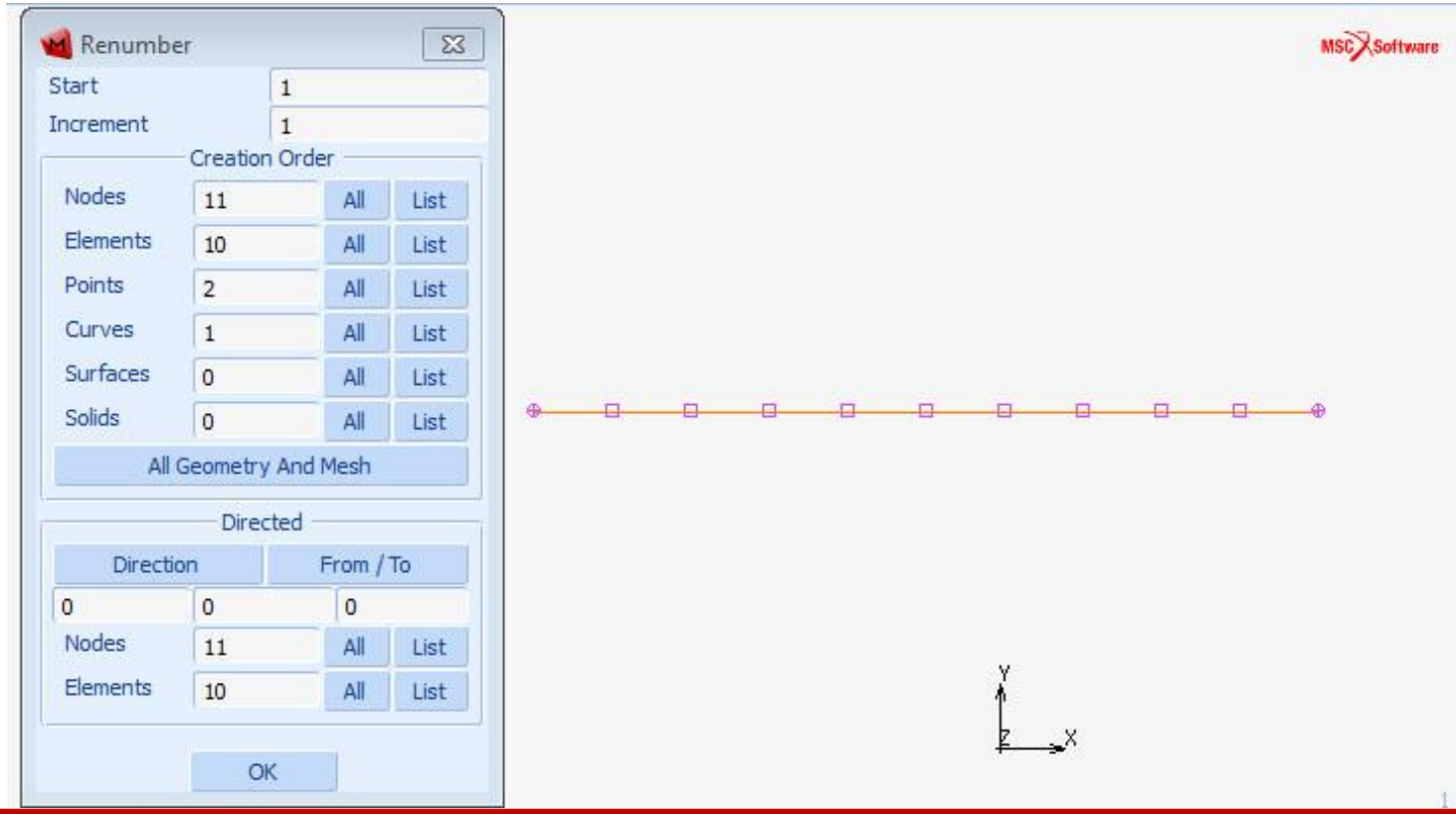
Creating 1D objects - example



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Creating 1D objects - example

CONGRATULATIONS!



You have just create your first mesh model!

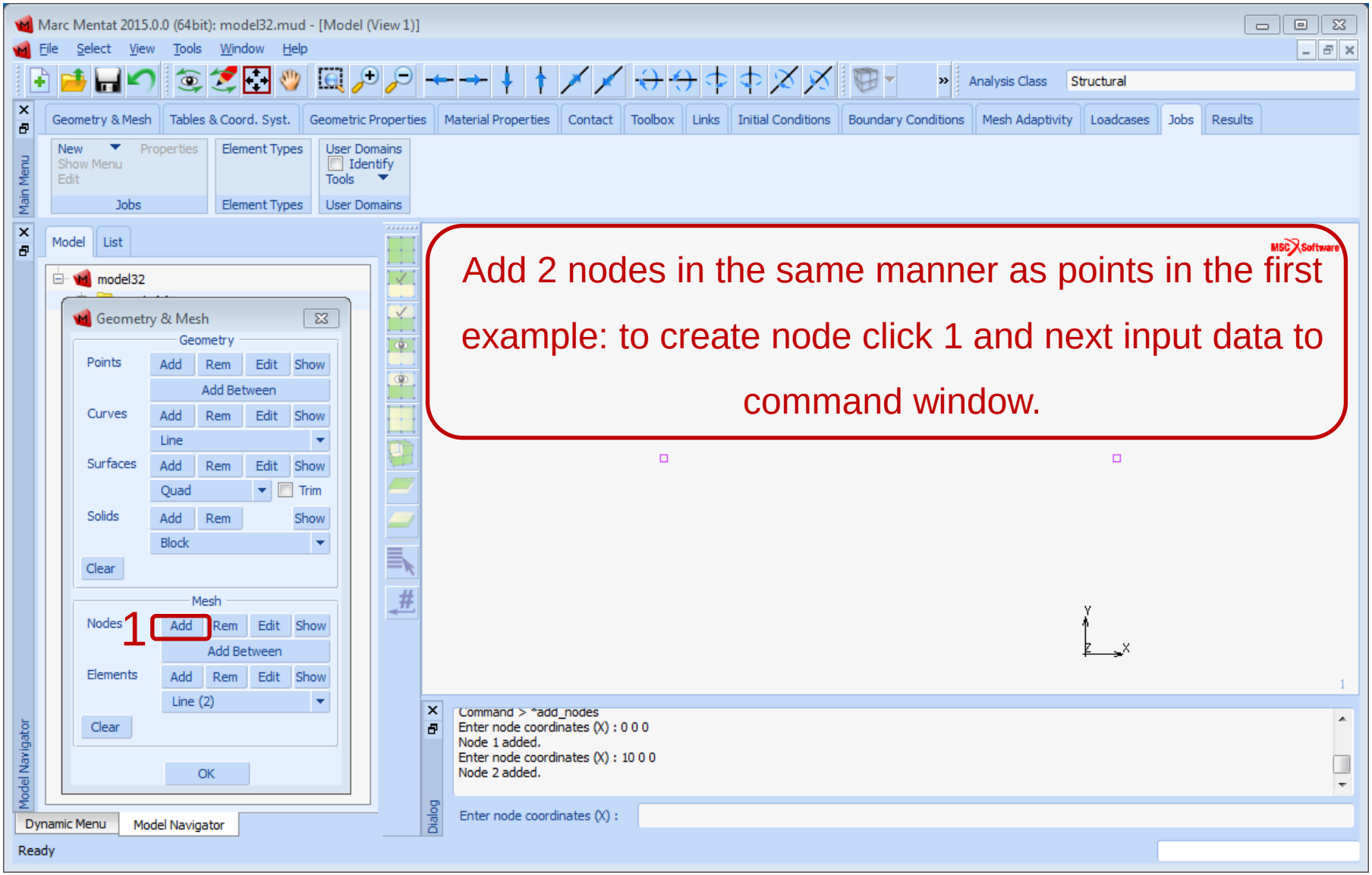
Pay attention that points are display as cross marker, while the nodes are display as square and circle markers.

1D modeling:

=> from geometry to mesh

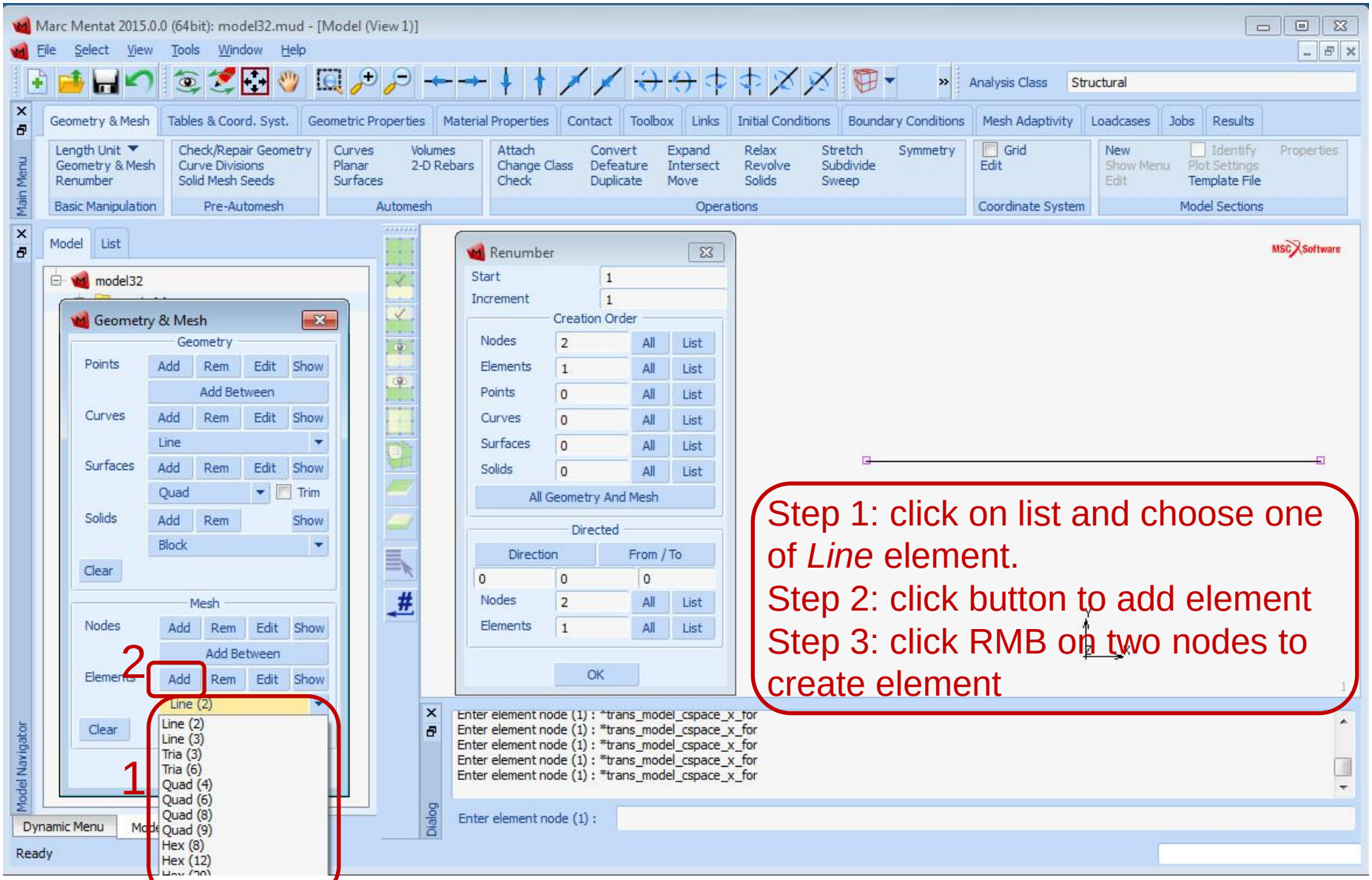
=> **direct mesh generation**

Creating 1D objects - example



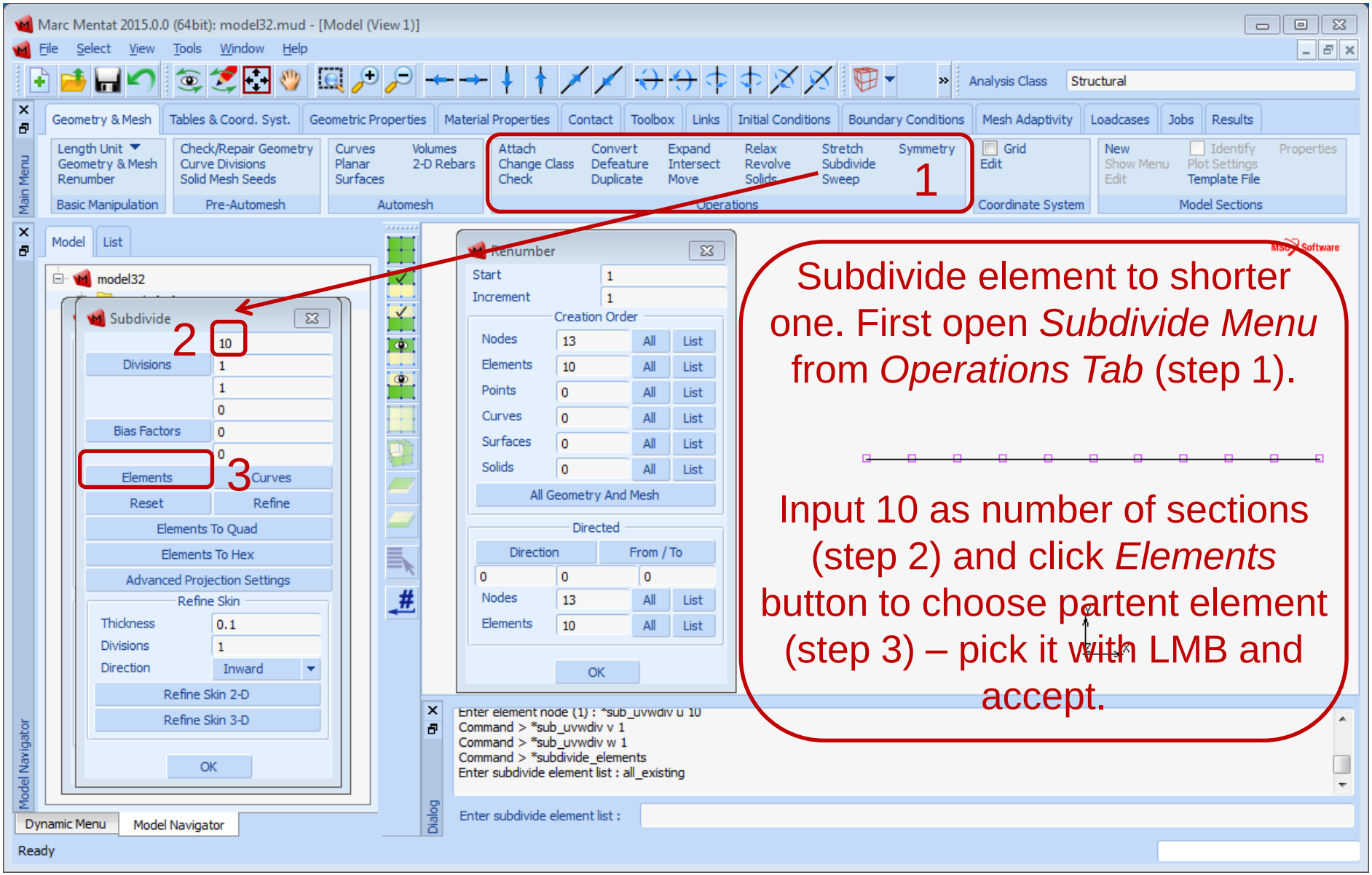
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Creating 1D objects - example



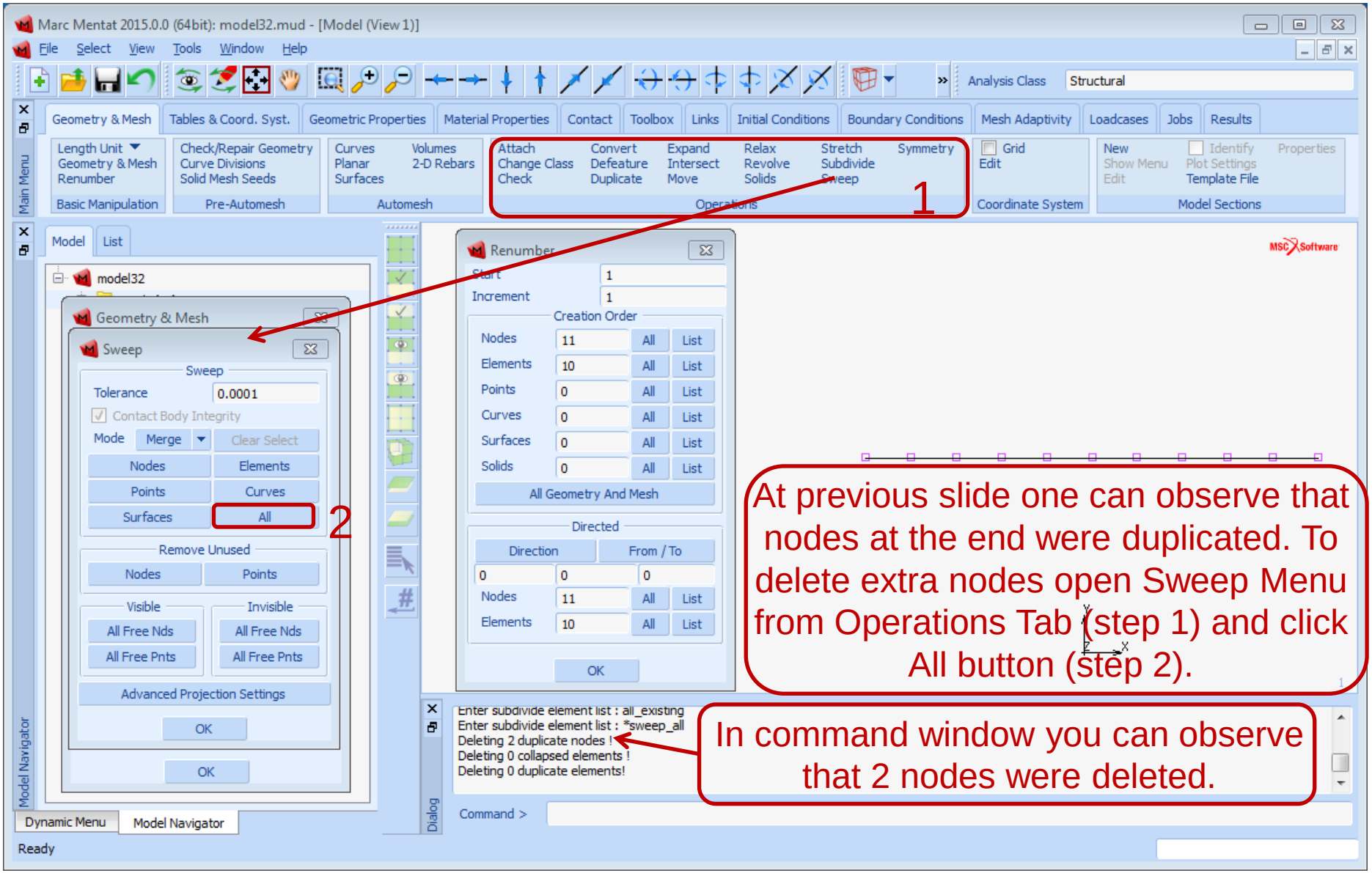
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Creating 1D objects - example



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Creating 1D objects - example



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2D modeling:

=> **from geometry to mesh**

=> **direct mesh generation**

The 2D objects are created in the same manner as 1D one.

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3D modeling:

=> ~~from geometry to mesh~~

=> **direct mesh generation**

=> import geometry and mesh

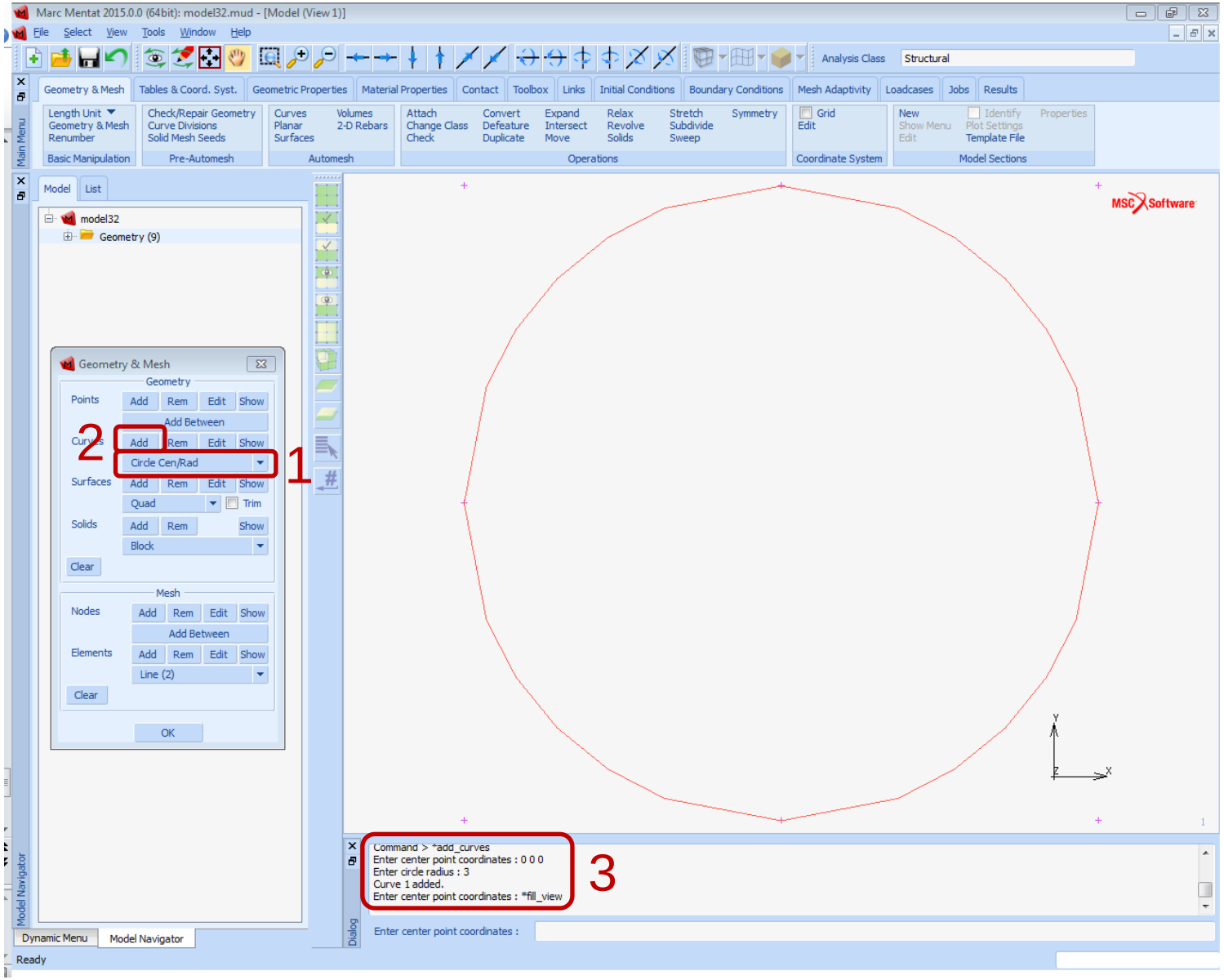
Creating volume object is unavailable at present license.
Therefore we start with direct mesh generation method.

Creating 3D objects – example 1

Step 1:
Choose type of Curve to Circle Cen/Rad

Step 2:
Click *Add Button* to create curve

Step 3:
Input data to command window about center point coordinates and radius value



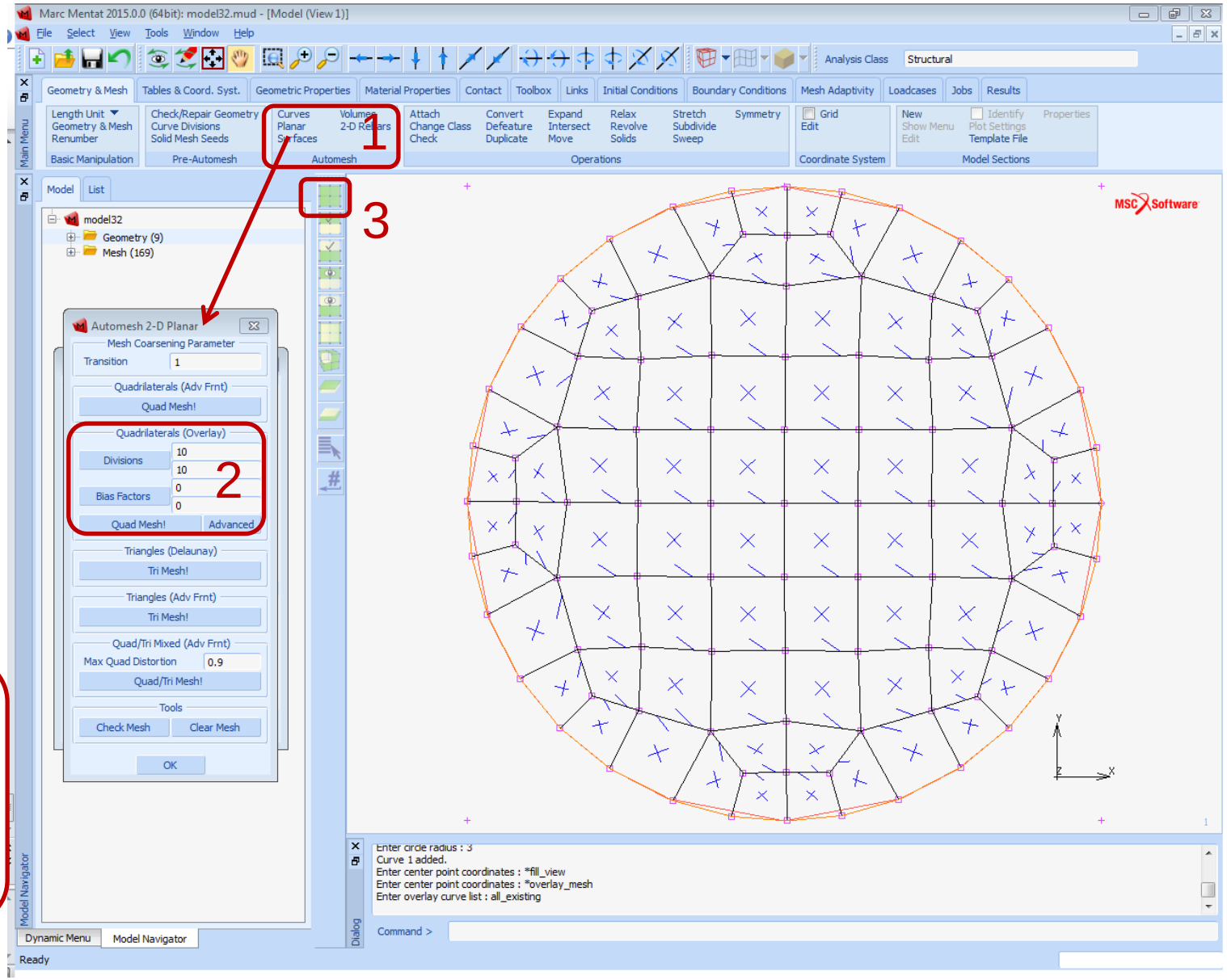
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Creating 3D objects – example 1

Step 1:
Open *Automesh 2-D Planar Menu* from *Automesh Tab*.

Step 2:
Input Divisions parameters and click *Quad Mesh!* Button

Step 3:
Choose curve – in this case we can choose all curves and so on clicking marked button



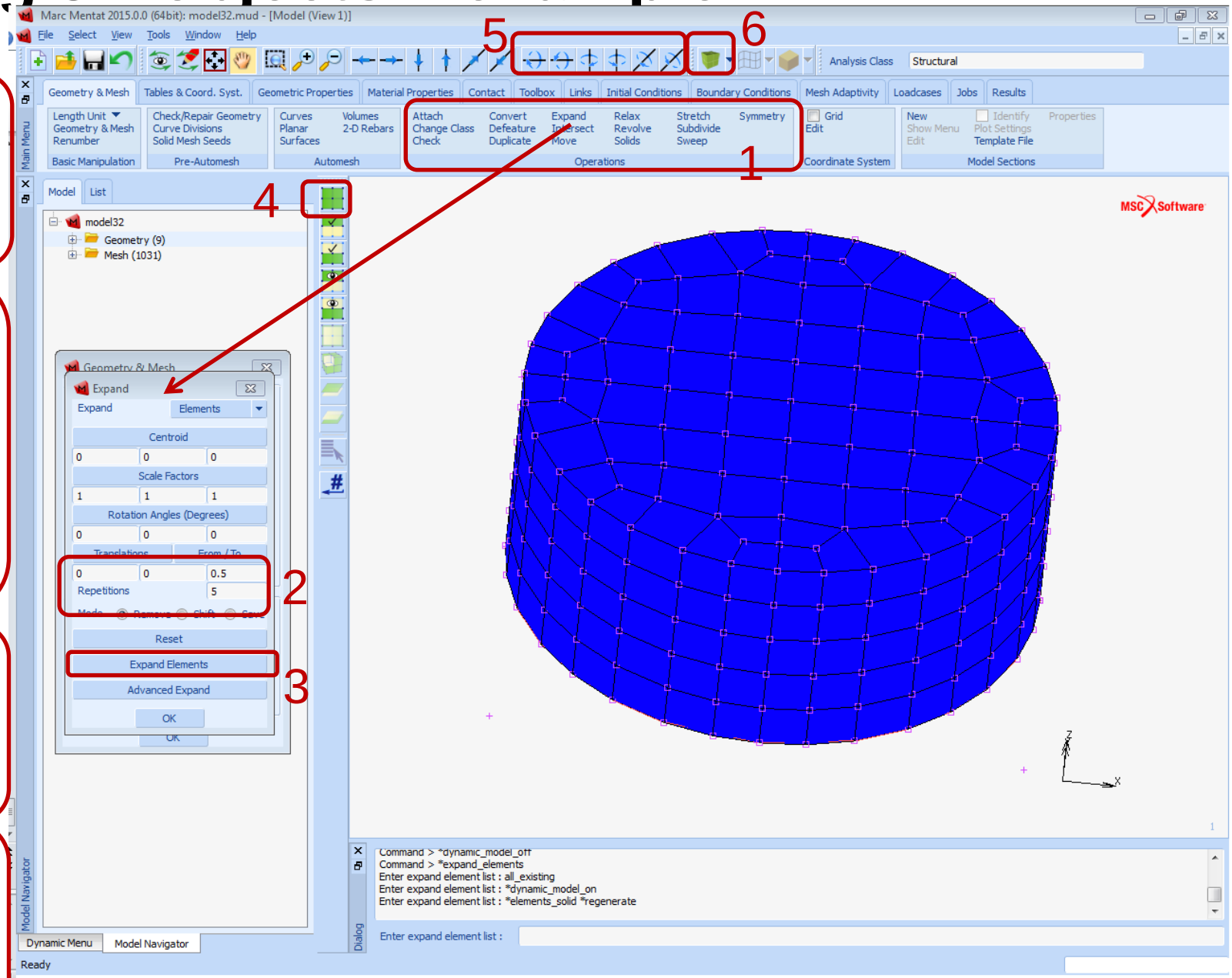
Creating 3D objects – example 1

Step 1:
Open *Expand Menu* from *Operations Tab*.

Step 2:
Input data about expand dimension and repetitions – this is number of element along expanded axis.

Step 3 & 4:
Click Expand Elements Button and choose all elements.

Step 5 & 6:
Rotate model and change view to solid to observe new Finite Elements.



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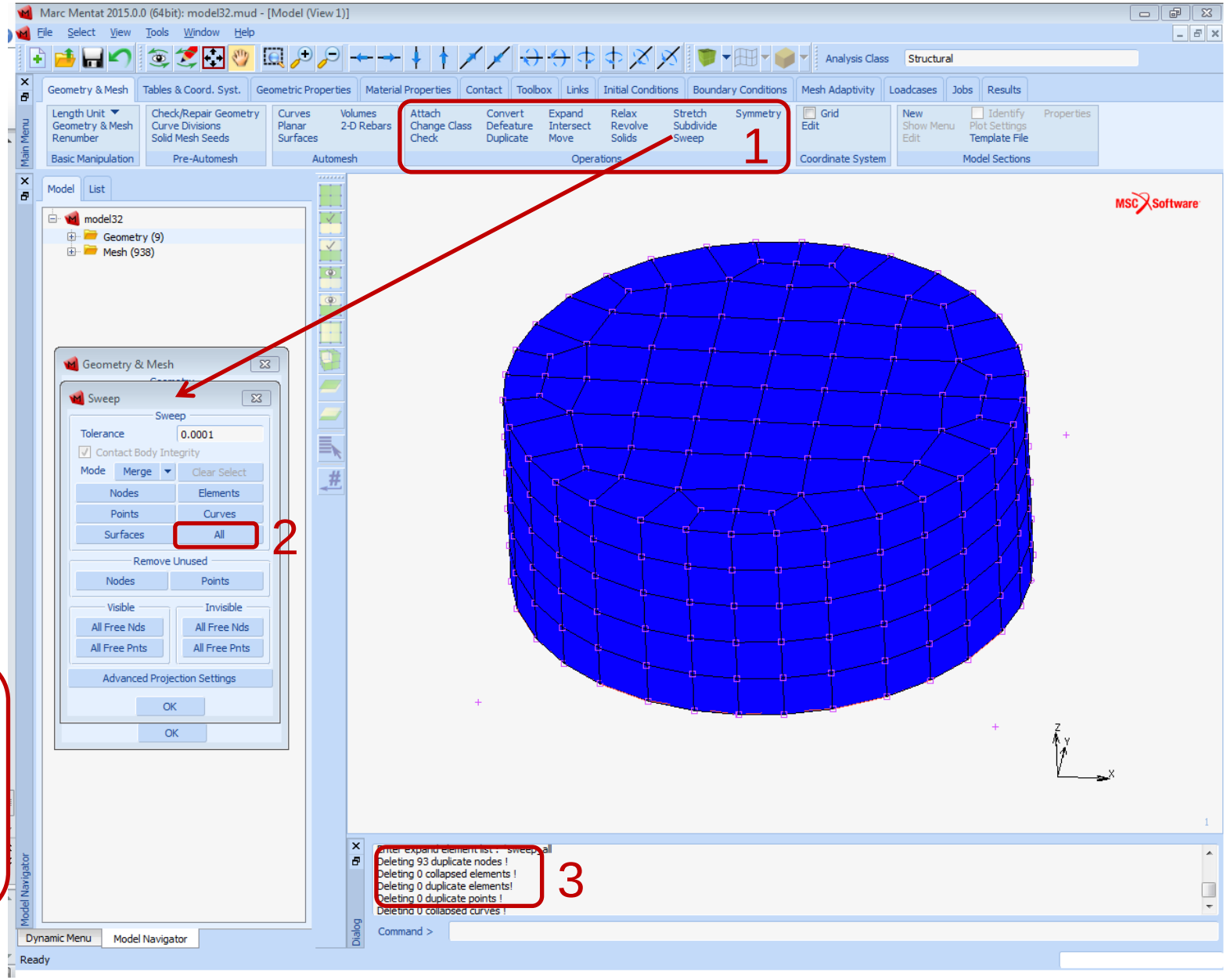
Creating 3D objects – example 1

Step 1:
Open *Sweep* Menu from *Operations* Tab.

Step 2:
Click *All* Button.

Step 3:
Observe results: 93 duplicated nodes deleted!

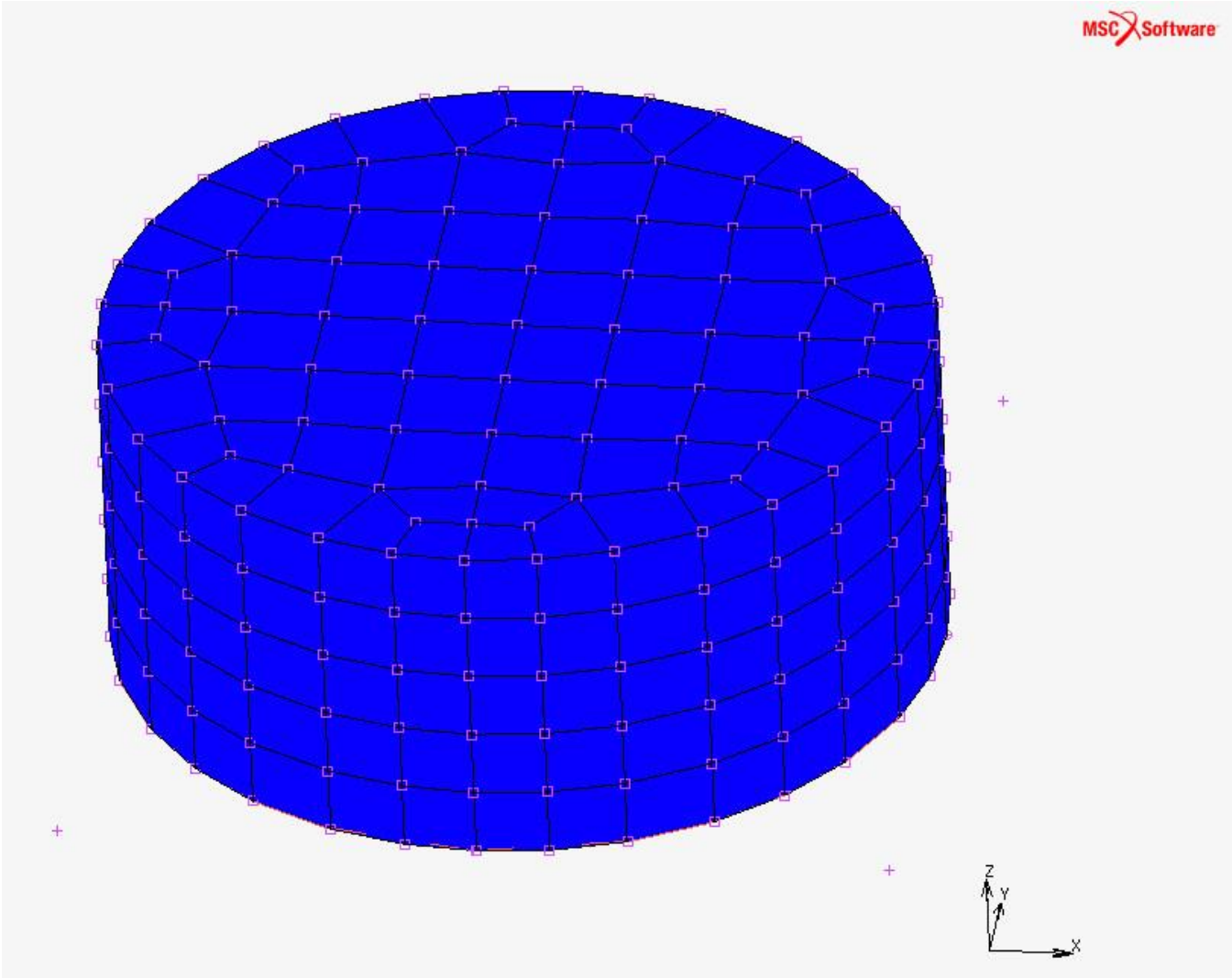
The sweep should be always done after FE generating!



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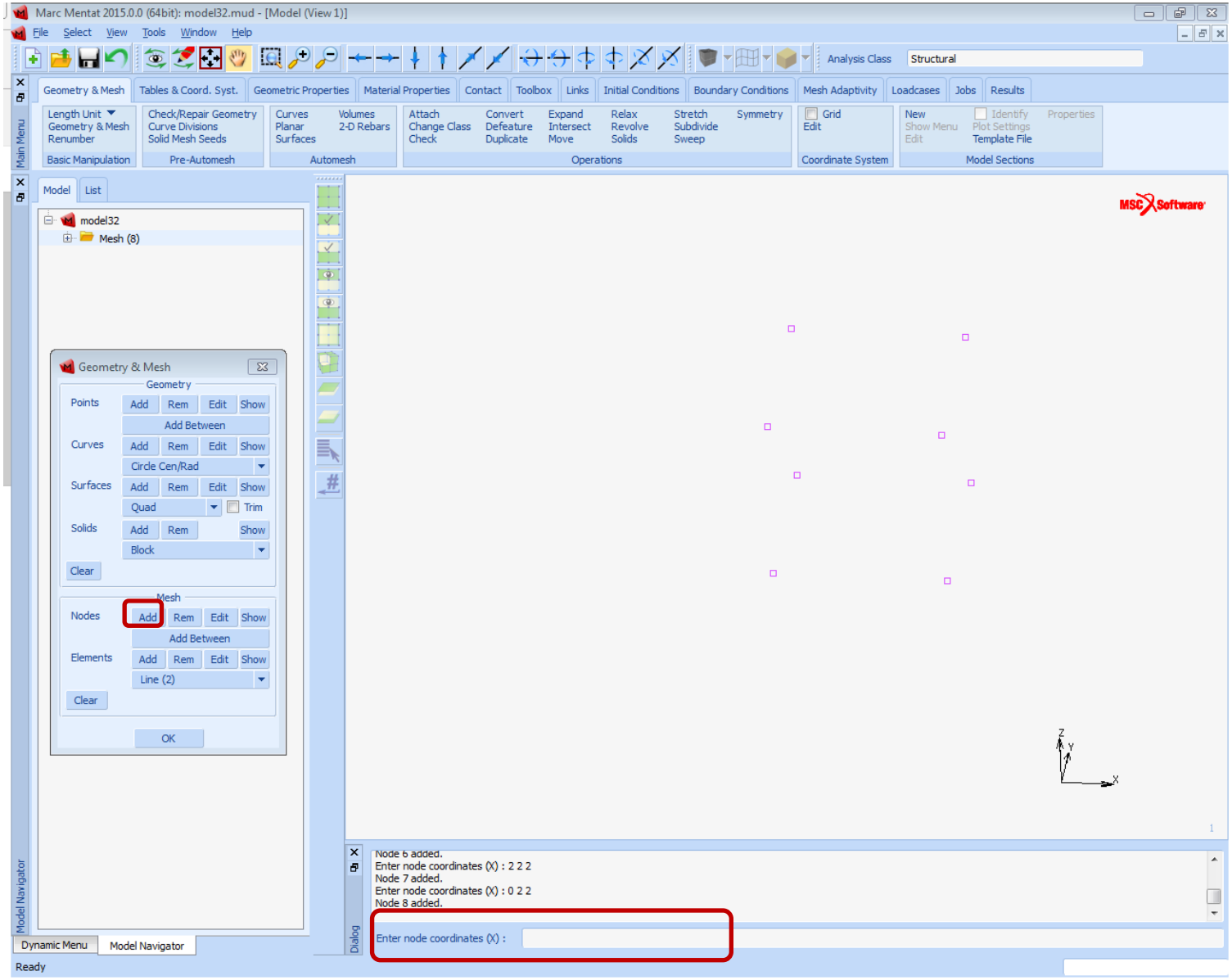
Creating 3D objects – example 1

CONGRATULATIONS!



Creating 3D objects – example 2

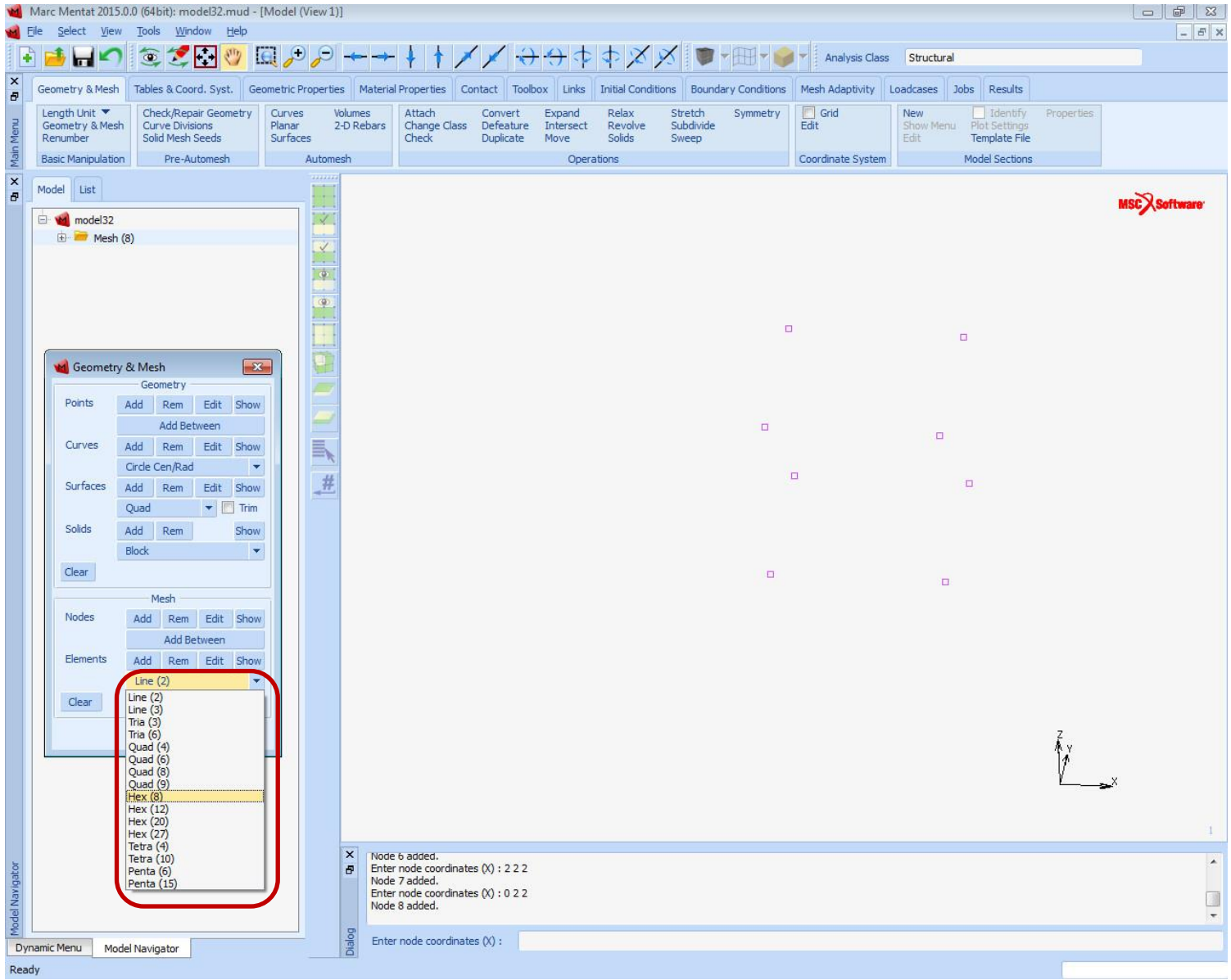
Step:
Add 8 nodes with coordinates:
-> 0 0 0
-> 2 0 0
-> 2 2 0
-> 0 2 0
-> 0 0 2
-> 2 0 2
-> 2 2 2
-> 0 2 2



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Creating 3D objects – example 2

Step:
Choose 3D Finite Elements with 8 nodes (cubic)

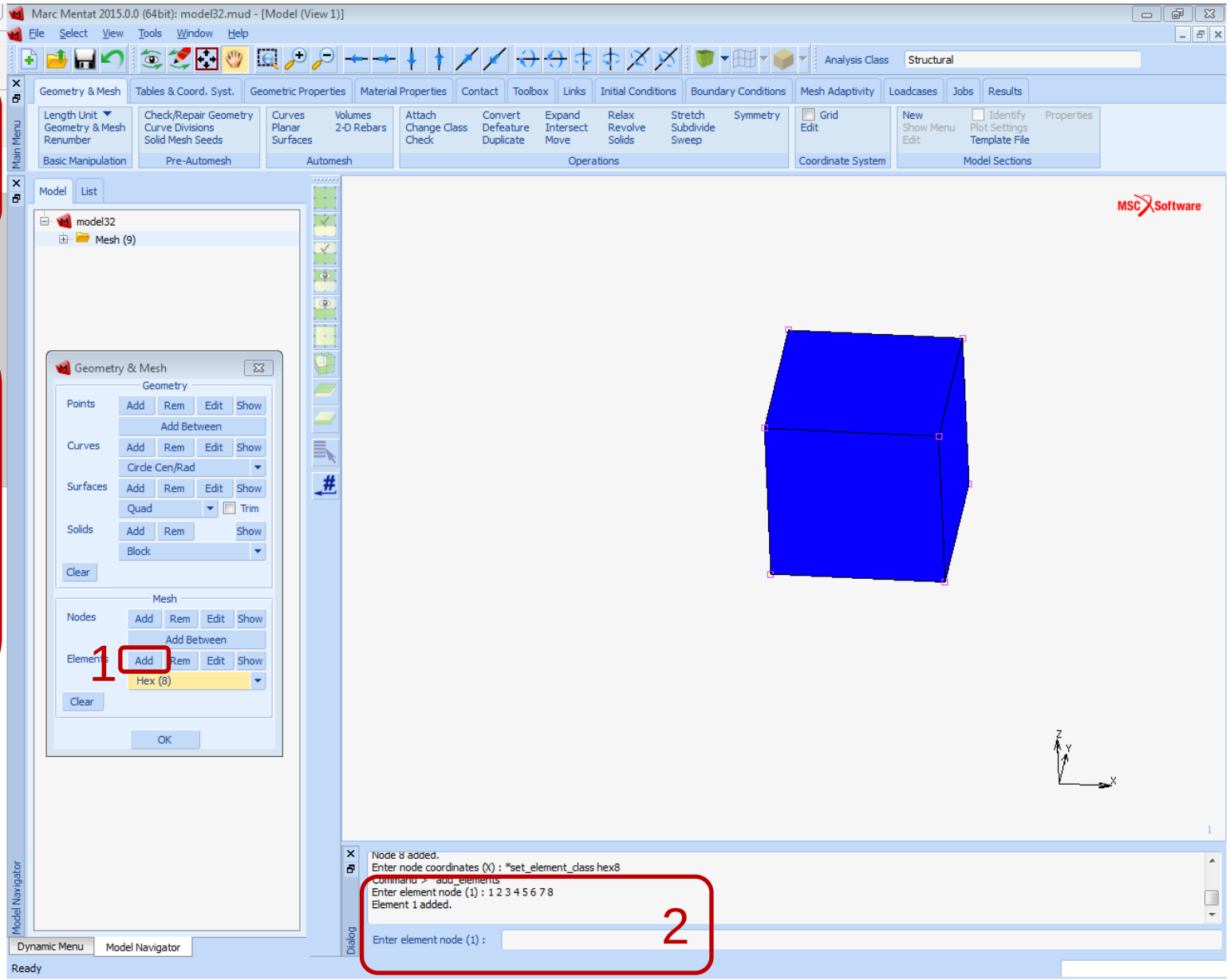


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Creating 3D objects – example 2

Step 1:
Click 1 (*Add* Button) to add Finite Element

Step 2:
Input indexes of nodes from which FE will be create to command window. In this case indexes of nodes will be: 1 2 3 4 5 6 7 8.



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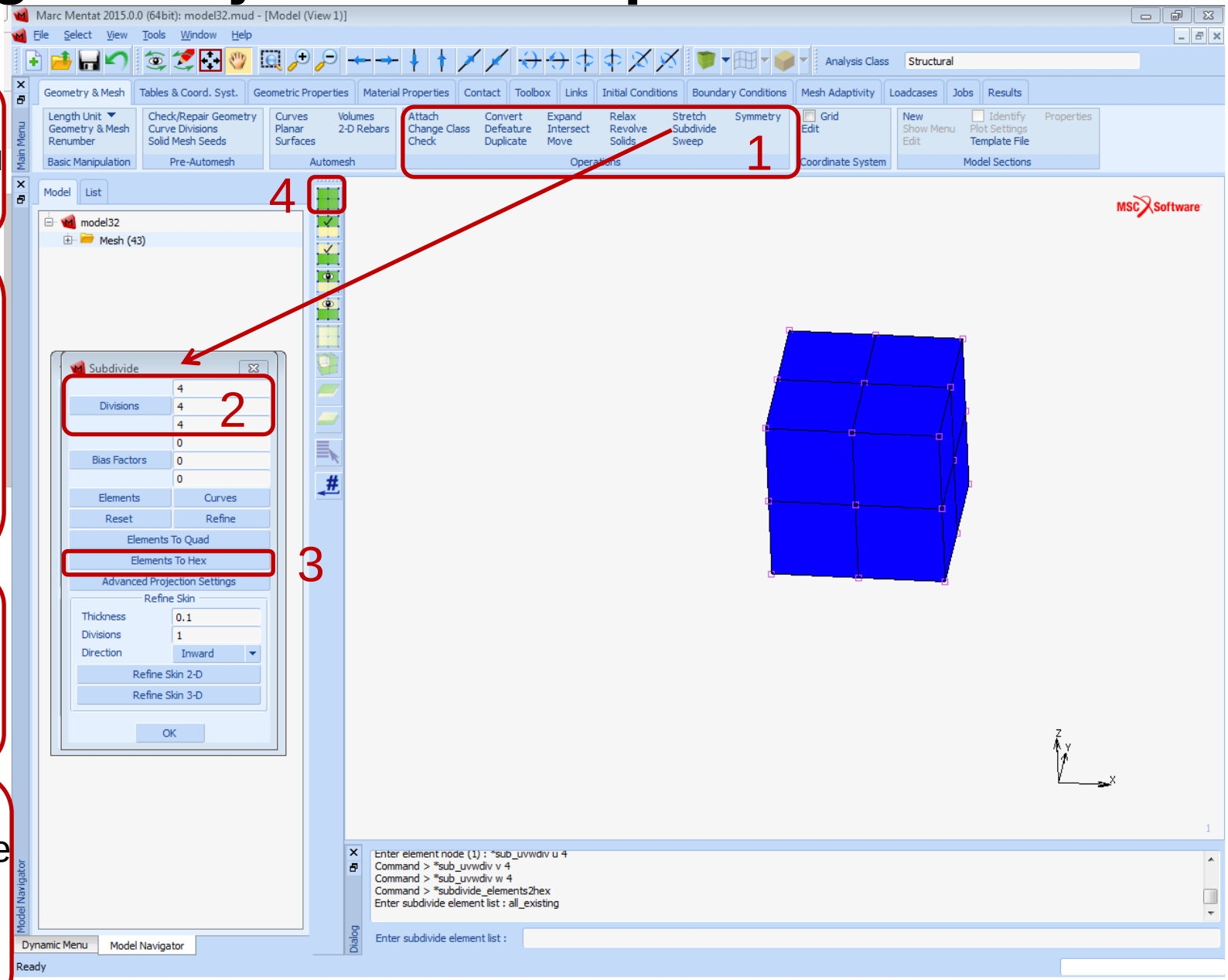
Creating 3D objects – example 2

Step 1:
Open *Subdivide* Menu from *Operations* Tab.

Step 2:
Input demanded values to *Divisions* parameters – accordingly for each axes

Step 3:
Click *Elements To Hex* Button to divide parent FE.

Step 4:
Choose FE which will be divided. Because there is only one FE you can choose all existing.



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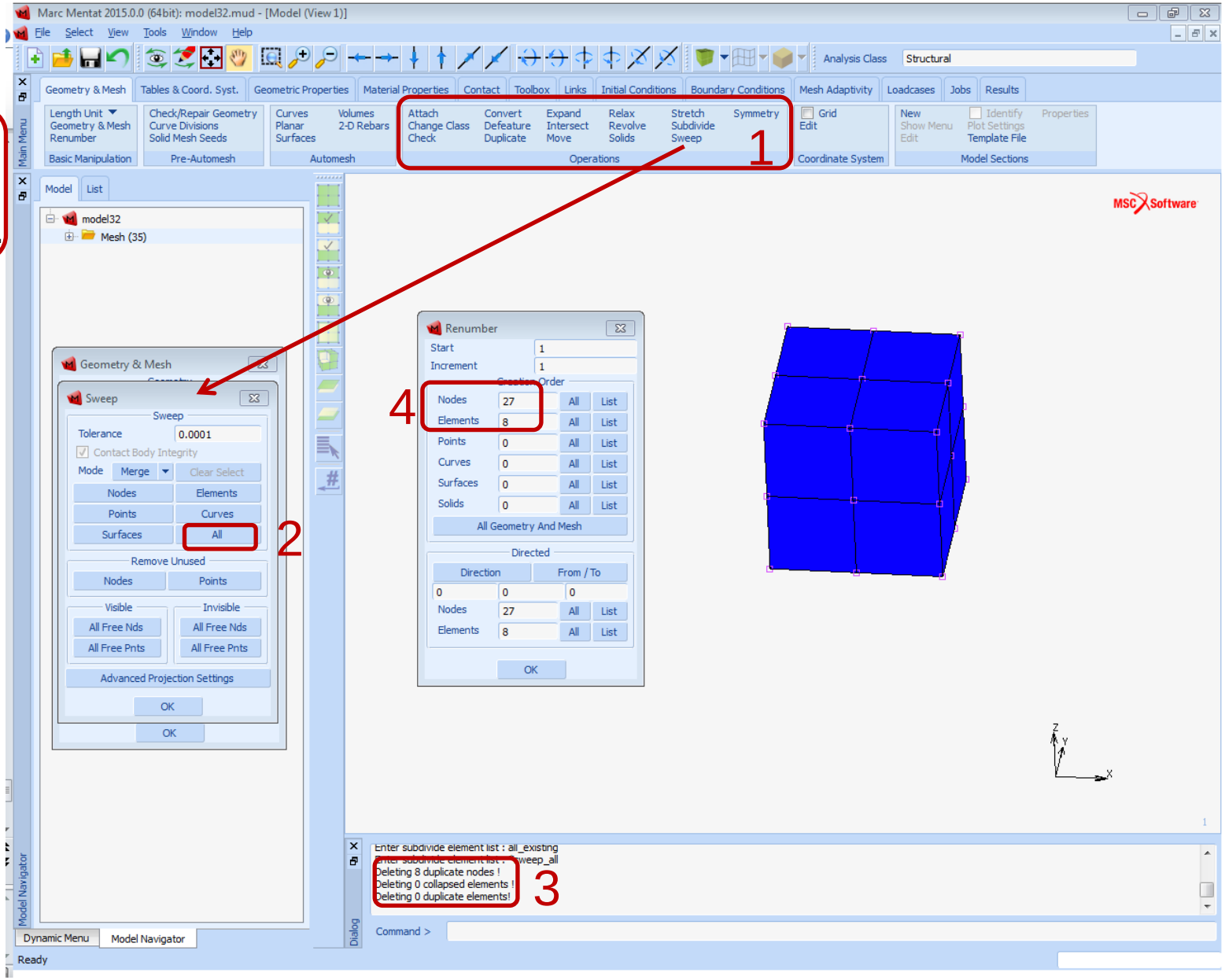
Creating 3D objects – example 2

Step 1:
Open *Sweep* Menu from *Operations* Tab.

Step 2:
Click *All* Buttons to delete duplicate items.

Step 3:
Observe results: 8 duplicated nodes deleted!

Step 4:
Verify number of created nodes and FEs.



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3D model:

=> direct mesh generation

=> **import geometry and mesh**

SOON….